

Mobility edges in off-diagonal disordered tight-binding models

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We study one-dimensional tight-binding models with a slowly varying, incommensurate off-diagonal modulation on the hopping amplitude. Interestingly, we find that the mobility edges can appear only when the off-diagonal (hopping) disorder is included, which is different from the known results induced by the diagonal disorder. We further study the situation where the off-diagonal and diagonal disorder terms (the incommensurate potential) are both included and find that the locations of mobility edges change significantly and the variation of the mobility edge becomes nonsmooth. To obtain these results, we first identify the exact expressions of mobility edges of both models by using asymptotic heuristic arguments, and then verify the conclusions by utilizing several numerical diagnostic techniques, including the inverse participation ratio, the density of states, and the Lyapunov exponent. These results will provide perspectives for future investigations on the mobility edge in low dimensional correlated disordered systems.

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I. INTRODUCTION

Anderson localization [1], a phenomenon in which destructive interference prevents wave propagation in a disordered medium, is an active research subject in condensed matter physics. The effect of the spatial dimension on Anderson localization is significant. The scaling theory [2] predicts that there is no localization transition in one- and two-dimensional systems where all wave functions, at least in the absence of interactions, are exponentially localized no matter how weak the strength of the uncorrelated disorder is. Compared with low dimensional systems, the three-dimensional (3D) Anderson model is very unique because the transition can occur at a finite disorder strength and there exists an energy threshold separating the localized and extended energy levels. Although this universal behavior is now well understood, a direct experimental observation of the critical energy, dubbed the mobility edge [3], remains a challenge due to the difficulty in realizing the 3D uncorrelated disorder. Thus, low dimensional correlated disorder systems attract a lot of research interest. With the development of ultracold atomic experiments, an optical speckle disorder potential can be realized by projecting a laser beam through a ground glass, and recently the mobility edge trajectory has been determined in the speckle disordered system with sufficiently high energy resolution [4–6].

Low dimensional quasiperiodic systems [7–14], which are considered as having highly correlated disorder, can also host mobility edges. As an important paradigm, the Aubry-André (AA) model [15], a 1D tight-binding model with an incommensurate potential, has a self-dual symmetry and can undergo a localization transition from the extended to localized phases depending on the relative strength of the incommensurate potential and the hopping amplitude. A very recent work [16,17] experimentally investigated the

localization properties of a bichromatic incommensurate lattice (a sketchy AA model) and demonstrated that there exists a mobility edge separating the localized and extended states in the intermediate phase. In the theoretical aspect, a unique class of systems with very slowly varying incommensurate potential in real space was revealed by Das Sarma *et al.* [18]. This deterministic (diagonal) potential is neither random nor simply incommensurate, and there are two mobility edges located at the spectrum where the eigenstates at the band center are all extended whereas the band-edge states are all localized. Later, different variations [19–25] of the 1D AA model containing mobility edges were studied. By including a long-range hopping term or constructing special forms of the on-site incommensurate potential, Refs. [26,27] demonstrated the existence of a mobility edge in a certain class of quasiperiodic systems which can be precisely addressed by self-dual symmetry.

In general, mobility edges in these models are all induced by the diagonal disorder. In this paper, we introduce a class of off-diagonal disordered tight-binding models, which are different from the previous diagonal disordered models since the generating mechanism of mobility edges in the spectrum significantly changes. Our generalized AA model, including a slowly varying, incommensurate off-diagonal modulation on the hopping term, is expressed as

$$\hat{H} = - \sum_{i=1}^{L-1} (t + \lambda_i) (\hat{c}_i^\dagger \hat{c}_{i+1} + \text{H.c.}) + \sum_{i=1}^L \mu_i \hat{n}_i, \quad (1)$$

where $\hat{c}_i$ ($\hat{c}_i^\dagger$) is the fermionic annihilation (creation) operator, $\hat{n}_i = \hat{c}_i^\dagger \hat{c}_i$ is the particle number operator, L is the total number of sites, $\lambda_i = \lambda \cos(2\pi\alpha i^v + \phi_\lambda)$ with $0 < v < 1$ and $\lambda > 0$ being the strength of the incommensurate modulation on the off-diagonal hopping amplitude, and $\mu_i = \mu \cos(2\pi\beta i^v + \phi_\mu)$ with $0 < v < 1$ and $\mu \geq 0$ being the strength of the on-site potential. Without loss of generality, we choose the

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parameters $\alpha = (\sqrt{5} - 1)/2$, the phase in the modulation $\phi_\lambda = \phi_\mu = 0$, and $v = 0.5$. For convenience, $t = 1$ is set as the energy unit. In particular, when $\beta = 0$ or $\beta = (\sqrt{5} - 1)/2$, the on-site potential μ_i becomes a constant or slowly varying incommensurate term, respectively. The total number of sites is set to be $L = 10\,000$ hereafter in this paper. In the next section, we will investigate both situations, and show that the locations of mobility edges in the presence of constant and slowly varying incommensurate potentials are completely different.

The rest of the paper is organized as follows. In Sec. II, we theoretically give the semianalytical arguments for the Hamiltonian (1). In Sec. III, we present our numerical results and compare them with the theoretical analysis. The conclusion is summarized in Sec. IV.

II. SEMIANALYTICAL METHODS

The slowly varying incommensurate modulation in 1D system brings new perspectives of the localization phenomenon due to its highly correlated disorder feature. To demonstrate the existence of mobility edges in the Das Sarma model [18], researchers developed an asymptotic semiclassical Wentzel-Kramers-Brillouin technique to calculate the density of states and the Lyapunov exponent, and they found that these two physical quantities are not necessarily smooth when crossing the mobility edge, which is substantially different from the 3D Anderson model. Here we present some preliminary heuristic arguments to obtain the explicit expressions of mobility edges semianalytically, then we verify our predictions by utilizing several typical numerical techniques.

By noticing that the difference of the off-diagonal, slowly varying incommensurate modulation $\lambda_i = \lambda \cos(2\pi\alpha i^v)$ vanishes in the thermodynamic limit [18], we thus write

$$\frac{d\lambda_i}{di} = -2\lambda\pi\alpha i^{v-1} \sin(2\pi\alpha i^v). \quad (2)$$

When $i \rightarrow \infty$, Eq. (2) can be written as

$$\lim_{i \rightarrow \infty} \left| \frac{d\lambda_i}{di} \right| = -\lim_{i \rightarrow \infty} 2\lambda\pi\alpha \frac{|\sin(2\pi\alpha i^v)|}{i^{1-v}} = 0, \quad (3)$$

since $0 < v < 1$. Equivalently, the modulation difference $\lambda_{i+1} - \lambda_i$ approaches 0 when the lattice number i is large enough, i.e., λ_i becomes a constant. This asymptotic property of “being constant” of λ_i is crucial for the localization property of our model.

When $\beta = 0$, μ_i becomes a constant μ and Hamiltonian (1) can be rewritten as the tight-binding Schrödinger equation

$$(E - \mu)\psi_m + (1 + C)\psi_{m-1} + (1 + C)\psi_{m+1} = 0, \quad (4)$$

where m is an arbitrary positive integer and $C = \lambda \cos[2\pi\alpha(i-1)^v] = \lambda \cos[2\pi\alpha(i+1)^v]$ since all the λ_i are constants in the large i limit. From Eq. (4) we obtain

$$\psi_{m+1} + \frac{E - \mu}{1 + C} \psi_m + \psi_{m-1} = 0. \quad (5)$$

Following the asymptotic heuristic argument [18], we write $\psi_m \sim Z^m$, where Z is a complex quantity. Then Eq. (5)

becomes

$$Z^2 + \frac{E - \mu}{1 + C} Z + 1 = 0, \quad (6)$$

to which the complex solutions are

$$Z_{1,2} = \frac{-B \pm \sqrt{B^2 - 4}}{2}, \quad (7)$$

with $B = \frac{E - \mu}{1 + C}$. From Eq. (7) we conclude that the amplitude Z is complex/extended if $|B| < 2$ whereas it is real/localized if $|B| > 2$. Noting that $(1 + C)_{\min} = 1 - \lambda$, due to the fact that μ is a constant number, we can incorporate μ into E and get

$$B_{\max} = \frac{|E - \mu|}{1 - \lambda}. \quad (8)$$

Here we emphasize that the crucial difference from the Das Sarma model is that only the off-diagonal term is disordered in this model, and the diagonal term μ is kept a genuine constant and can be incorporated into E directly in the semianalytical method, whereas in the Das Sarma model only the diagonal term μ is disordered (not a genuine constant) and cannot be incorporated into E directly. Therefore, the conditions for extended and localized solutions are respectively given by

$$B_{\max} < 2 \Rightarrow -2(1 - \lambda) + \mu < E < 2(1 - \lambda) + \mu$$

(extended),

$$B_{\max} > 2 \Rightarrow E < -2(1 - \lambda) + \mu \cup 2(1 - \lambda) + \mu < E$$

(localized). (9)

Note that for the existence of the mobility edges there is an implicit condition $1 - \lambda > 0$. If this condition is satisfied, there will be two mobility edges denoted by $E_{\pm c1} = \pm 2(1 - \lambda) + \mu$. Hence, the model defined by the Hamiltonian (1) with $\beta = 0$ has extended states at the band center ($E_{-c1} < E < E_{+c1}$) and localized states at the band edges ($E > E_{+c1}$, $E < E_{-c1}$). Interestingly, even if μ becomes very large, the mobility edges $E_{\pm c1}$ still exist, as shown in Fig. 1. This is a new feature of the model, with off-diagonal slowly varying incommensurate modulation, whereas the previous models with diagonal slowly varying incommensurate modulation always drive the system to an all-wave-function-localized transition point and the mobility edge vanishes accordingly.

When $\beta = (\sqrt{5} - 1)/2$, the system becomes more complicated. In this case, μ_i denotes the slowly varying incommensurate potential, thus there exist competing off-diagonal and diagonal incommensurate terms in Hamiltonian (1), which can be written as

$$(E - C')\psi_m + (1 + C)\psi_{m-1} + (1 + C)\psi_{m+1} = 0, \quad (10)$$

where $C' = \mu \cos(2\pi\beta i^v)$ and other parameters are the same as those in Eq. (4). Following the same procedure, we can obtain $B' = \frac{E - C'}{1 + C}$. Note $(1 + C)_{\min} = 1 - \lambda$ and $(E - C')_{\max} = |E + \mu|$; then we have

$$B'_{\max} = \frac{|E + \mu|}{1 - \lambda}, \quad (11)$$

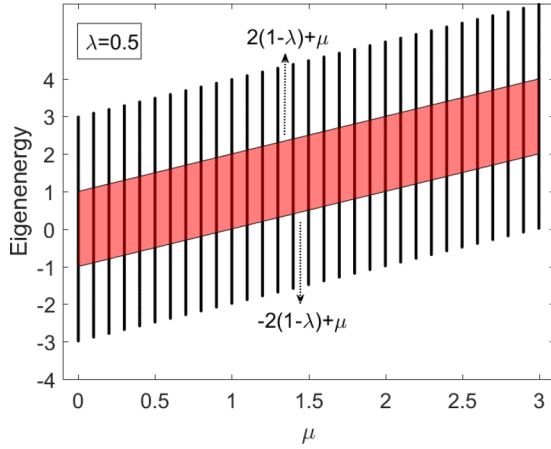


FIG. 1. Eigenenergy of Eq. (1) as a function of μ with $\lambda = 0.5$ when $\beta = 0$. The black curves denote the eigenenergies of the system and the curves covered by the red quadrilateral correspond to the extended states, whereas uncovered curves correspond to the localized states. The upper and lower solid lines of the red quadrilateral represent two mobility edges denoted by $E_{\pm c1} = \pm 2(1 - \lambda) + \mu$. This model does not have an all-wave-function-localized phase transition point.

Thus, the conditions for extended and localized solutions are respectively given by

$$\begin{aligned}
 B'_{\max} < 2 &\Rightarrow -2(1 - \lambda) - \mu < E < 2(1 - \lambda) - \mu \\
 &\quad \text{(extended),} \\
 B'_{\max} > 2 &\Rightarrow E < -2(1 - \lambda) - \mu \cup 2(1 - \lambda) - \mu < E \\
 &\quad \text{(localized).} \quad (12)
 \end{aligned}$$

However, the result given by Eq. (12) is imperfect. In fact, it is only applicable for $0 < \mu < 2\lambda$ by numerical verification, while for $2\lambda < \mu < 2$ the upper mobility edge $2(1 - \lambda) - \mu$ remains unchanged and the lower mobility edge becomes $-2(1 + \lambda) + \mu$. Thus, the varying trend of the mobility edge becomes nonsmooth at the specific point $\mu = 2\lambda$. We note that a similar singular effect has been predicted [28] in the 3D speckle disorder model due to the peculiarity of the potential correlation function. However, Ref. [29] points out that this singularity is smoothed out in the exact numerical calculation. For this model, the nonsmooth change of the mobility edge is definitive.

We have not found an exact analysis for these mobility edges. Here we provide a naive picture to explain this phenomenon. We can make a proper deduction that when the strength of the on-site potential μ_i becomes very large, all wave functions of the system must be localized and the mobility edges must vanish due to the on-site potential being incommensurate. When the system reaches the all-wave-function-localized transition point, the two mobility edges must intersect at one point. However, the results given by Eq. (12) never intersect. Therefore, with the amplitude increments of μ_i , the slope of the lower mobility edge must change sign, i.e., $-2(1 - \lambda) - \mu \Rightarrow -2(1 - \lambda) + \mu$, if the upper mobility edge remains unchanged. However, this change is discontinuous, and the continuously varying condition of the lower mobility edge requires $-2(1 - \lambda) + \mu \Rightarrow -2(1 + \lambda) + \mu$. Thus, we

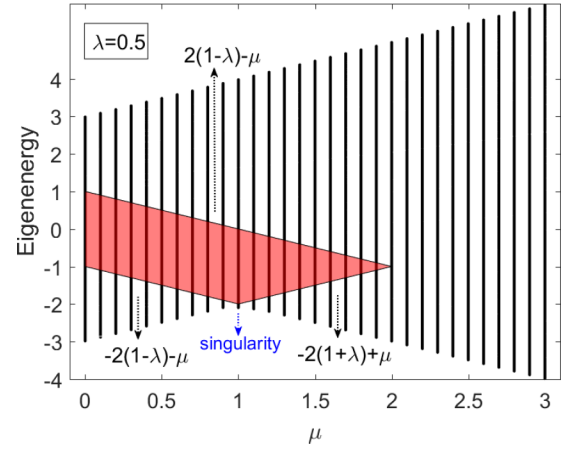


FIG. 2. Eigenenergy of Eq. (1) as a function of μ with $\lambda = 0.5$ when $\beta = (\sqrt{5} - 1)/2$. The black curves denote the eigenenergies of the system and the curves covered by the red quadrilateral correspond to the extended states, whereas uncovered curves correspond to the localized states. The upper and lower solid lines of the red quadrilateral represent two mobility edges denoted by $E_{\pm c2} = \pm 2(1 - \lambda) - \mu$ ($0 \leq \mu \leq 2\lambda$) or $E_{+c2} = 2(1 - \lambda) - \mu$ and $E_{-c2} = -2(1 + \lambda) + \mu$ ($2\lambda < \mu < 2$). The blue arrow signals the nonsmooth singularity point $\mu = 2\lambda$. In this model the all-wave-function-localized phase transition point is located at $\mu = 2$.

can get $\mu = 2\lambda$ from $-2(1 - \lambda) - \mu = -2(1 + \lambda) + \mu$ and $\mu = 2$ from $2(1 - \lambda) - \mu = -2(1 + \lambda) + \mu$, and two mobility edges intersect at $\mu = 2$, as shown in Fig. 2. In addition, for the existence of the mobility edges there is also an implicit condition $2\lambda < 2$, i.e., $\lambda < 1$. We summarize the expressions of mobility edges as follows:

$$E_{\pm c2} = \begin{cases} \pm 2(1 - \lambda) - \mu & (0 \leq \mu \leq 2\lambda), \\ 2(1 - \lambda) - \mu, -2(1 + \lambda) + \mu & (2\lambda < \mu < 2). \end{cases} \quad (13)$$

Thus, the model defined by the Hamiltonian (1) with $\beta = (\sqrt{5} - 1)/2$ has extended states at the band center ($E_{-c2} < E < E_{+c2}$) and localized states at the band edges ($E > E_{+c2}$, $E < E_{-c2}$), and the all-wave-function-localized transition point is located at $\mu = 2$.

III. NUMERICAL VERIFICATION

To support the semianalytical results given in the previous section, we now present detailed numerical calculation. We diagonalize the model Hamiltonian (1) directly to get the eigenenergies E and the associated wave functions ψ . Then, typical physical quantities used in the disordered system, such as the inverse participation ratio, the density of states, and the Lyapunov exponent, can be obtained to distinguish the localized and extended states in the spectrum.

First we calculate the inverse participation ratio (IPR). The IPR of a normalized wave function ψ is defined as [30–32]

$$\text{IPR}_n = \sum_{j=1}^L |\psi_j^n|^4, \quad (14)$$

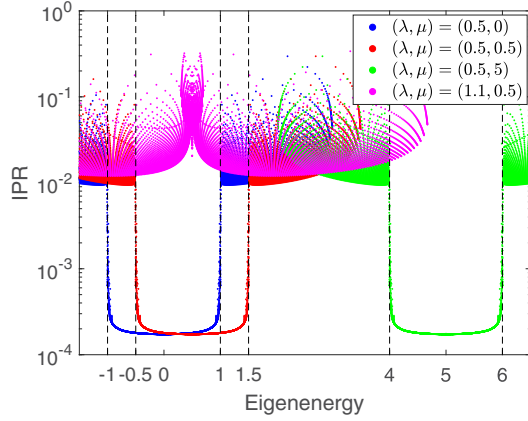


FIG. 3. The distribution of IPR as a function of eigenenergy for various (λ, μ) when $\beta = 0$. Black dotted lines correspond to two turning points of IPR located at the mobility edges $E_{\pm c1} = \pm 2(1 - \lambda) + \mu$. When $(\lambda, \mu) = (0.5, 0)$, $(0.5, 0.5)$, and $(0.5, 5)$, $E_{+c1} = 1, 1.5, 6$ and $E_{-c1} = -1, -0.5, 4$ are located at the spectrum respectively, while when $(\lambda, \mu) = (1.1, 0.5)$ there are no mobility edges and all wave functions are localized due to $\lambda = 1.1 > 1$.

where L denotes the total number of sites and n is the index of energy level. It is well known that the IPR of the extended state scales like L^{-1} , which approaches 0 in the thermodynamic limit but is finite for a localized state.

Figures 3 and 4 plot the IPR of the corresponding wave functions as a function of the eigenenergy for various (λ, μ) when $\beta = 0$ and $(\sqrt{5} - 1)/2$, respectively. We find that, as the eigenenergy varies, the IPR changes dramatically from the order of magnitude 10^{-2} (a typical value for the localized states) to 10^{-4} (a typical value for the extended states) or inversely at certain energies. This jumping phenomenon of IPR indicates that there exist mobility edges in the energy spectrum.

If $\lambda = 1.1$, the IPR of all wave functions in both cases are of the order of magnitude of 10^{-2} and localized. None of them appear around 10^{-4} or are extended, and this verifies the implicit condition $\lambda < 1$ for the existence of mobility edges. Remarkably, in Fig. 3, even if the strength of the potential ($\mu = 5$ in this situation) becomes very large, the mobility edges $E_{\pm c1}$ still exist. In Fig. 4, when the strength of the slowly varying incommensurate potential is larger than the threshold ($\mu = 2$), all wave functions are localized and there are no mobility edges in the spectrum. As a result, a metal-insulator transition appears at $\mu = 2$ if $\beta = (\sqrt{5} - 1)/2$, which is different from the $\beta = 0$ case. We implement calculations for various (λ, μ) and find that these mobility edges are exactly located at $E_{\pm c1}$ and $E_{\pm c2}$, respectively, as analyzed in Sec. II.

To strengthen the validity of the results of IPR, we also calculate the density of states $D(E)$ and the Lyapunov exponent $\gamma(E)$ of this quasiperiodic system. Here $D(E)$ is defined as

$$D(E) = \sum_{n=1}^L \delta(E - E_n) \quad (15)$$

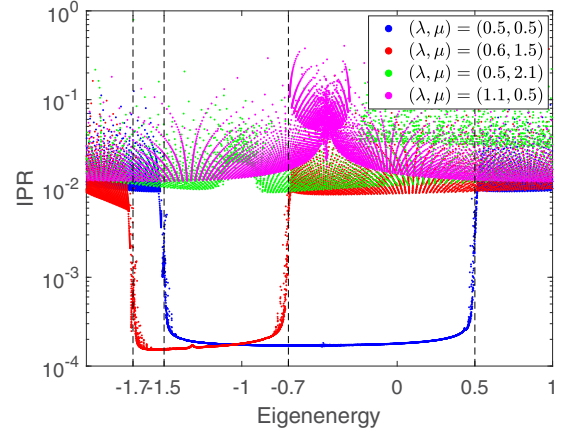


FIG. 4. The distribution of IPR as a function of eigenenergy for various (λ, μ) when $\beta = (\sqrt{5} - 1)/2$. Black dotted lines correspond to two turning points of IPR located at the mobility edges denoted by $E_{\pm c2} = \pm 2(1 - \lambda) - \mu$ if $0 \leq \mu \leq 2\lambda$ or by $E_{+c2} = 2(1 - \lambda) - \mu$ and $E_{-c2} = -2(1 + \lambda) + \mu$ if $2\lambda < \mu < 2$. When $(\lambda, \mu) = (0.5, 0.5)$ and $(0.6, 1.5)$, $E_{+c2} = 0.5, -0.7$ and $E_{-c2} = -1.5, -1.7$ respectively are located at the spectrum, while when $(\lambda, \mu) = (0.5, 2.1)$ and $(1.1, 0.5)$, there are no mobility edges and all wave functions are localized due to $\mu = 2.1 > 2$ and $\lambda = 1.1 > 1$, respectively.

and $\gamma(E)$ [18] is

$$\gamma(E_n) = \frac{1}{L-1} \sum_{n \neq m}^L \ln |E_n - E_m|, \quad (16)$$

where E_n is the n th eigenenergy and L is the total number of sites. The Lyapunov exponent is defined as the inverse localization length, hence $\gamma = 0$ for an extended state whereas $\gamma \neq 0$ for a localized state. The density of states and the Lyapunov exponent are connected by the relation

$$\gamma(E) = \int dE' D(E') \ln |E - E'|. \quad (17)$$

In Fig. 5 we plot the density of states (DOS) and the Lyapunov exponent γ as a function of the eigenenergy when $\beta = 0$. To avoid losing generality, we choose three different sets of parameters with $(\lambda, \mu) = (0.5, 0)$, $(0.5, 0.5)$, and $(0.5, 5)$. In Fig. 5(a) it is obvious that the DOS in our model displays sharp peaks at certain energies, which is different from the 3D Anderson random disorder [33,34]. The reason is due to the nature of the slowly varying incommensurate modulation [18], and this peak-like singularity of the DOS reflects the change of the nature of the eigenstates. Therefore, the extended-localized transitions corresponding to two mobility edges $E_{\pm c1}$ are indicated by two sharp peaks of the DOS. In Fig. 5(b) γ is plotted for the same sets of parameters; it also exhibits singular behaviors at two mobility edges, just as those for the DOS.

In Fig. 6 we show the numerical results for the DOS [Fig. 6(a)] and γ [Fig. 6(b)] when $\beta = (\sqrt{5} - 1)/2$. They also exhibit similar singularity as the eigenenergy passes through the two mobility edges $E_{\pm c2}$ for $\mu < 2$. When $(\lambda, \mu) = (0.5, 2.1)$, the DOS and γ vary smoothly and do not exhibit any singular behavior, so there exist no mobility edges and all

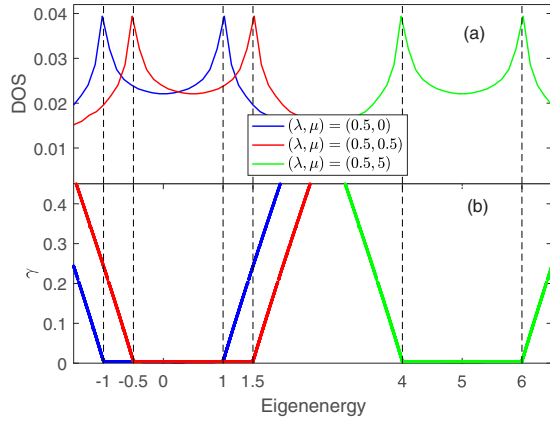


FIG. 5. DOS and the Lyapunov exponent γ as a function of eigenenergy for various (λ, μ) when $\beta = 0$. (a) Obviously a dramatic change of DOS occurs when the eigenenergy passes through the mobility edges $E_{\pm c1}$, which are in accordance with the IPR predictions. (b) When the eigenenergy is located in the interval $[E_{-c1}, E_{+c1}]$, $\gamma \rightarrow 0$, indicating that the corresponding state is extended. Otherwise γ is finite, indicating that the corresponding state is localized.

wave functions are localized compared to the $\mu < 2$ case; this agrees with the semianalytical prediction given in Sec. II. We have also checked other combinations of parameters and we get the same results, as expected.

Another interesting subject is the critical exponents of the Lyapunov exponent at the mobility edge, defined by

$$\gamma(E) \sim |E - E_{\pm ci}|^{\theta}, \quad i = 1, 2, \quad (18)$$

in the localized regions of energy spectrum. Similarly, the behavior of the density of states at the mobility edge can be

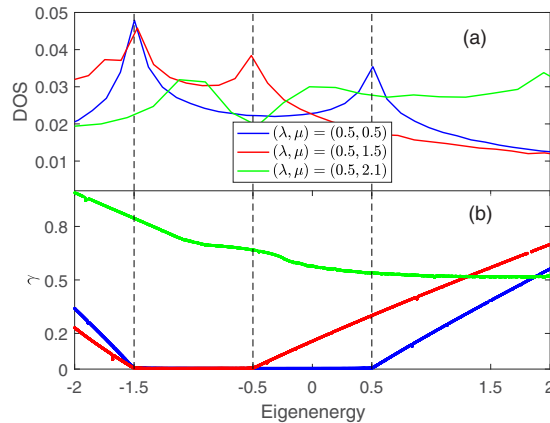


FIG. 6. DOS and the Lyapunov exponent γ as a function of eigenenergy for various (λ, μ) when $\beta = (\sqrt{5} - 1)/2$. (a) Obviously a dramatic change of DOS occurs when the eigenenergy passes through the mobility edges $E_{\pm c2}$ when $\mu < 2$, while when $\mu = 2.1 > 2$ there is no obvious change. (b) When $\mu < 2$, $\gamma \rightarrow 0$ for the eigenenergy located in the interval $[E_{-c2}, E_{+c2}]$, indicating that the corresponding state is extended, otherwise γ is finite, indicating that the corresponding state is localized. When $\mu = 2.1 > 2$, the γ is finite as the eigenenergy varies, indicating that all wave functions are localized.

written as

$$D(E) \sim |E - E_{\pm ci}|^{-\delta}, \quad i = 1, 2. \quad (19)$$

Therefore, according to Eq. (17) the critical exponents θ and δ are clearly related by the equation

$$\theta + \delta = 1. \quad (20)$$

In Figs. 5(b) and 6(b), we can identify that γ is linear with the eigenenergy varying in the localized region, i.e., the band edges ($E > E_{+c1,2}$, $E < E_{-c1,2}$), which leads to the fact that $\theta = 1$ and $\delta = 0$. These results agree with those of the known model [18], and the parameters λ , μ , and v are all found to be irrelevant to the critical exponents θ and δ . In addition, we also find that these mobility edges depend on λ and μ but are irrelevant to v by varying the parameters.

In the AA model, the Lyapunov exponent is known analytically to be $\gamma = \ln(V/2)$, where V is the strength of the AA incommensurate potential. Thus, near the phase transition point $V_c = 2$, $\gamma \sim |V - V_c|^\theta$, where the critical exponent θ is known to be exactly 1. In Ref. [35], this critical exponent was numerically reproduced to a reasonably high accuracy, and the authors also calculated the critical exponent of the generalized Aubry-André (GAA) model with mobility edges, and found that it is 1 too when crossing the mobility edge. It is particularly worth noting that the GAA model is still self-dual under a generalized duality transformation [27]. Furthermore, the authors pointed out that $\theta = 1$ strongly violates the Chayes-Chayes-Fisher-Spencer (CCFS) bound $\theta \geq 2/d$ (d is the spatial dimension) for a stable disorder-driven quantum critical point. This is because the CCFS bound is obtained for random disorder, which is not suitable for the incommensurate models since there is no intrinsic disorder. This provides a major distinction between the Anderson localization driven by the destructive interference and the AA-type localizations originating from the spectral property of the almost periodic Schrödinger operator.

Regarding to the slowly varying incommensurate model, the critical exponents θ of the Das Sarma model (the diagonal disorder) and our model (the off-diagonal disorder) are both 1. These results strongly corroborate the analysis of Ref. [35] that the critical exponents of the AA-type localizations belong to a unique class. Though the critical exponents near the transition points of these incommensurate models are the same, we must emphasize that the Lyapunov exponents γ of both AA and GAA models increase logarithmically, whereas those of both Das Sarma's model and our model increase linearly. This also signifies that the localization nature of the slowly varying incommensurate models is not exactly the same as that of the incommensurate models with self-duality.

IV. CONCLUSIONS

In this work we have studied the localization properties of a class of the off-diagonal disordered tight-binding models with a constant on-site potential ($\beta = 0$) and a slowly varying incommensurate on-site potential [$\beta = (\sqrt{5} - 1)/2$]. We find the following interesting features of the two models and show that the semianalytical critical conditions of mobility edges are in excellent agreement with the localization properties obtained from the numerical calculation.

(1) When $\beta = 0$, we reveal that there exist two mobility edges $E_{\pm c1} = \pm 2(1 - \lambda) + \mu$ separating the localized and extended states in the spectrum. More interestingly, this phenomenon completely results from the off-diagonal disorder term, and even if the strength of the on-site potential μ becomes very large, the mobility edges $E_{\pm c1}$ still exist.

(2) When $\beta = (\sqrt{5} - 1)/2$, there also exist two mobility edges $E_{\pm c2}$ in the spectrum. However, when the incommensurate potential strength satisfies $0 \leq \mu \leq 2\lambda$, we have $E_{\pm c2} = \pm 2(1 - \lambda) - \mu$. When $2\lambda < \mu < 2$, we have $E_{+c2} = 2(1 - \lambda) - \mu$ and $E_{-c2} = -2(1 + \lambda) + \mu$. When $\mu > 2$, all wave functions of the model are localized.

Finally, we would like to point out that bichromatic optical lattices have been realized in ultracold atomic systems by superimposing two standing waves with mutually incommensurate wave vectors. Thus, the AA localization is quantified by directly monitoring the transport of the noninteracting Bose-Einstein condensate [9]. The off-diagonal random disorder was also experimentally realized recently [36]. However, the

measurement of the mobility edge trajectory is a very hard task. This is mainly due to the insufficient energy resolution; i.e., DOS in these disordered models varies smoothly. Also, implementing the experiment requires a sharp maximum of DOS around the mobility edge. However, for the slowly varying incommensurate models, DOS reaches maximum at the singular point (the mobility edge), so it is possible to locate the mobility edge with sufficient precision if this highly correlated disorder configuration can be realized. We believe that the interesting features of these models will bring new perspectives to a wide range of correlated disordered systems.

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